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COMP 009

Object Oriented Programming

Instructions: kindly read the questions carefully and write your answer in the yellow pad paper together with name, course, subject and signature on it.

1. What is Java programming language? Discuss its key features and advantages.

> Java is a high-level, object-oriented programming language that was developed by Sun microsystems and is currently owned by Oracle Corporation. It is a simple, secure, portable, and robust programming language. It is used to develop desktop applications, web-based applications, mobile applications, and large-scale enterprise-level applications. It is an object-oriented programming languge, which means that a programmer can use object-oriented programming to develop a program. Object-oriented programming makes a program more manageable, maintainable, and efficient.

Key features and advantages:

It supports object-oriented programming, which allows a programmer to use object-oriented programming to develop a program. Object-oriented programming provides several programming concepts, such as inheritance, polymorphism, encapsulation, and abstraction. It is a platform-independent programming languge, which means that a program written in java can run on any platform, It is a secure programming languge because a programmer cannot access memory directly. And this is also robust programming languge because of its memory management and exception handling.

2. Explain the concept of platform independence in Java. How does Java achieve platform independence?

> Platform independence in Java refers to the fact that a java program can run on any platform without changing the code. This is achieved in Java using bytecode and the java virtual machine (JVM). When a java program is compiled, it is not converted to machine code. Instead, it is converted to bytecode. This bytecode can run on any platform with the JVM

installed. the JVM converts this bytecode to manchine code for the operating system. Since each platform has its anywhere. This is the reason for platform independence in java. This concept in java can be summarized by the statement "Write Once, Run Anywhere!"

3. Discuss the main components of a Java program and their roles.

> The major parts of a java program are class, main method, statements, methods, and objects. A java program should have at least one class. A class is used to create objects in a program. the main method is the starting point in a program. Inside the main method, these are statements that are executed by the computer. A method is a set of instructions that perform a particular operation in a program. Objects are used to represent real-world entities in a program.

4. What are the differences between JDK, JRE, and JVM in Java?

> In java, JDK, JRE, and JVM have different uses, JDK stands for (Java development kit). JDK includes tools that are used to develop Java programs, such as the Java compiler.
JRE stands for (Java Runtime Environment). JRE is used to run java programs and includes the JVM.
JVM stands for (Java Virtual Machine). JVM is used to run java programs and make java platform-independent. In simple words, JDK is used by developers to develop Java programs, JRE is used by users to run java programs, and JVM is used to run the java bytecode.

5. Explain the difference between interpreted and compiled languages. Is java an interpreted or compiled language?

> Compiled programming languages are those in which the source code of the program is converted to machine code before it is executed. C and C++ programming languages fall under this category. on the other hand, interpreted programming languages are those in which the code is interpreted line by line, Python programming language uses interpretation. Java can be classified under compiled and interpreted programming languages. this is because it uses compilation to convert the source code to bytecode, it uses interpretation to execute the code.